ReVSI: Rebuilding Visual Spatial Intelligence Evaluation for Accurate Assessment of VLM 3D Reasoning

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Research Questions

- How can we fix systematic validity issues in VLM spatial intelligence evaluation where annotations from 3D reconstructions don't match what's visible in video frames?
- How does frame sampling rate affect the answerability and correctness of spatial reasoning questions in VLM evaluation?
- What are the true spatial reasoning capabilities of modern VLMs when evaluated under properly controlled conditions?

Core Idea

Existing VLM spatial intelligence benchmarks have fundamental validity problems: they use 3D reconstruction-based annotations as ground truth for video evaluation, but these annotations often miss visible objects or have wrong labels due to reconstruction artifacts. Additionally, they assume models see all frames while real VLMs use sparse sampling (16-64 frames), making many questions unanswerable. ReVSI fixes this by re-annotating everything based on actual video content and ensuring questions are answerable under the model's actual frame budget. The key insight is enforcing strict consistency between what the model sees and what the benchmark asks.

Key Formula

Mean Relative Accuracy (MRA)

$$MRA = \frac{1}{|C|}\sum_i \left(\frac{|\hat{y}-y|}{y} < 1-\theta\right)$$

C is confidence threshold set, y is ground truth, ŷ is prediction, θ is dynamic threshold. Penalizes large deviations while rewarding predictions close to ground-truth, used for numerical question evaluation.

Key Figure

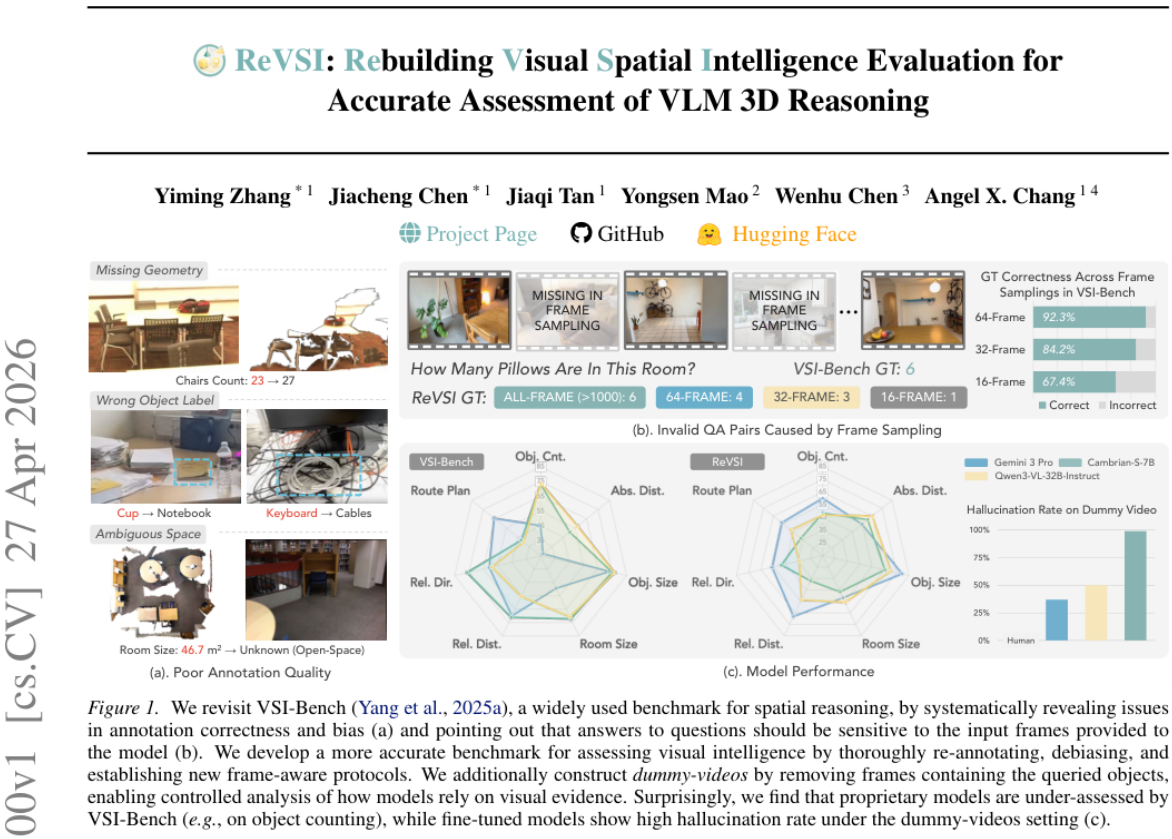

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Figure 1: Shows the core problem: VSI-Bench has annotation errors, frame sampling makes questions unanswerable, and models hallucinate on dummy videos without visual evidence.

Limitations

- Relies on costly expert-level human annotation which limits scalability to larger datasets
- High-quality 3D indoor spatial intelligence dataset limits applicability to outdoor or dynamic scenes
- Frame sampling analysis focused on uniform sampling may not reflect optimal sampling strategies
- Evaluation limited to English language spatial reasoning tasks